

- **Problem Statement ID** – 26181
- **Problem Statement Title** – A secure, AI-powered Personal Health Companion for real-time, privacy-preserving health monitoring and early warning, improving resilience during heat waves, floods, pollution events and other disasters through on-device intelligence
- **Theme** – MedTech / BioTech / HealthTech
- **PS Category** – Hardware
- **Team ID** – 157650
- **Team Name** – ABBOY



ForeverFit + ForeverBand: privacy-preserving on-device AI for real-time health monitoring and disaster early warning

An ESP32-S3 wrist wearable streams vitals, motion and environment over BLE to a phone app that detects health anomalies, falls and local hazards with edge AI, then raises alerts and SOS, even with no internet.

PROPOSED SOLUTION (idea / prototype)

- **ForeverBand:** ESP32-S3 battery-powered (500 mAh) wrist wearable with MAX30102 (heart rate, SpO2), DS18B20 (body temperature), MPU6050 (motion), BME280 (temperature, humidity, pressure) and an OLED watch face
- **ForeverFit:** Flutter Android app (plus a native iOS SwiftUI app) for continuous health monitoring, trends and alerts
- **On-device AI:** fall and activity CNNs, personal-baseline anomaly detection and an offline Gemma LLM assistant
- **Disaster early warning:** heat wave, AQI, flood, cyclone, earthquake and pressure-drop alerts on an offline risk map
- **Emergency assistance:** automatic fall / SOS escalation to emergency services and caregivers with vitals and location

DETAILED EXPLANATION OF THE PROPOSED SOLUTION

- 1 **Sense:** MAX30102 PPG at 100 Hz (HR / SpO2 reported only after a 2 s settle and 4 clean beats), environment 1 Hz, motion 20 Hz
- 2 **Stream:** BLE GATT notify (vitals, environment, motion) plus time-sync and watch settings
- 3 **Store:** local Hive database on the phone; nothing is uploaded
- 4 **Infer:** 3 s IMU windows go to the fall CNN; the activity CNN sets heart-rate limits; a 7-day baseline flags values more than 2 SD from normal
- 5 **Fuse:** vitals + NOAA heat index + AQI + hazards give a 0–100 wellness score
- 6 **Act:** fall or SOS starts a 10 s "I'm OK" countdown, then emergency call, then emergency contact (5 retries), then SMS
- 7 **Background:** a foreground service keeps fall and disaster checks running with the screen off

HOW IT ADDRESSES THE PROBLEM (PS 26181 Expected Solution coverage)

PS PILLAR	WHAT FOREVERFIT / FOREVERBAND DELIVERS	STATUS
1. Continuous Health Monitoring	HR, SpO2, body temp, activity, steps, sleep; 7-day personal baseline; wellness score	BUILT
2. AI Health Anomaly Detection	Activity-aware abnormal HR, low SpO2, fever / hypothermia, heat stress, all on-device	BUILT
3. Disaster-Specific Alerts	Heat-wave, AQI / respiratory, flood, cyclone and earthquake alerts; siren for imminent disasters	BUILT
4. Environmental Awareness	BME280 temp / humidity / pressure; 3 hPa-in-3 h pressure-drop warning; AQI, PM2.5, PM10	BUILT
5. Privacy-Preserving Edge AI	All analysis on-device, local storage, offline maps and LLM, user-controlled export	BUILT
6. Emergency Assistance	Automatic fall detection, manual SOS, caregiver call / SMS, location in the spoken summary	BUILT
7. Personal Wellness Dashboard	Daily summary, 7 / 30 / 90-day trends, wellness score, hydration and medication reminders	BUILT
8. Scalable Deployment	Android + iOS apps; open BLE protocol for other bands; consumers, providers, agencies	PARTIAL

INNOVATION AND UNIQUENESS OF THE SOLUTION

Offline on-device LLM

Gemma 4 E2B runs on the phone, grounded in the user's own vitals; no cloud chatbot

Evidence-tuned fall AI

Threshold chosen by sweeping held-out data and confirmed with real drop tests

Vitals + hazard fusion

Checks whether local conditions are dangerous for this user, not just sends a weather push

Context-aware thresholds

Heart-rate limits depend on activity and personal baseline, which cuts false alarms

Graceful degradation

Works on the phone alone; the wearable, internet and GPS are all optional

TECHNOLOGIES TO BE USED

HARDWARE AND FIRMWARE (FOREVERBAND)

ESP32-S3 N16R8 MAX30102 PPG HR / SpO2
DS18B20 body temp MPU6050 IMU BME280 env
SSD1306 OLED 500 mAh battery + 3.3 V regulator
C++ / Arduino NimBLE BLE GATT

MOBILE APPS

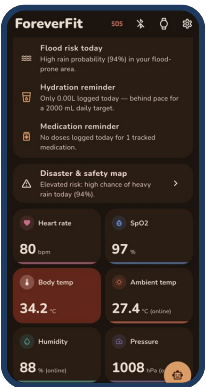
Flutter / Dart (Android) Swift / SwiftUI (iOS)
Hive (local DB) Kotlin telephony

ON-DEVICE AI / ML

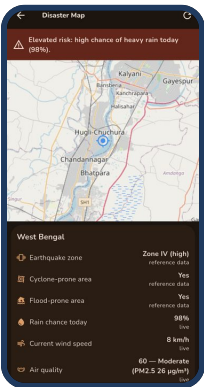
TensorFlow Lite (LiteRT) 1D-CNN
Gemma 4 E2B LLM flutter_gemma Python + Keras
UMAFall + MotionSense

CONNECTIVITY AND BACKGROUND

flutter_blue_plus geolocator offline map tiles
foreground service



Dashboard



Disaster risk map

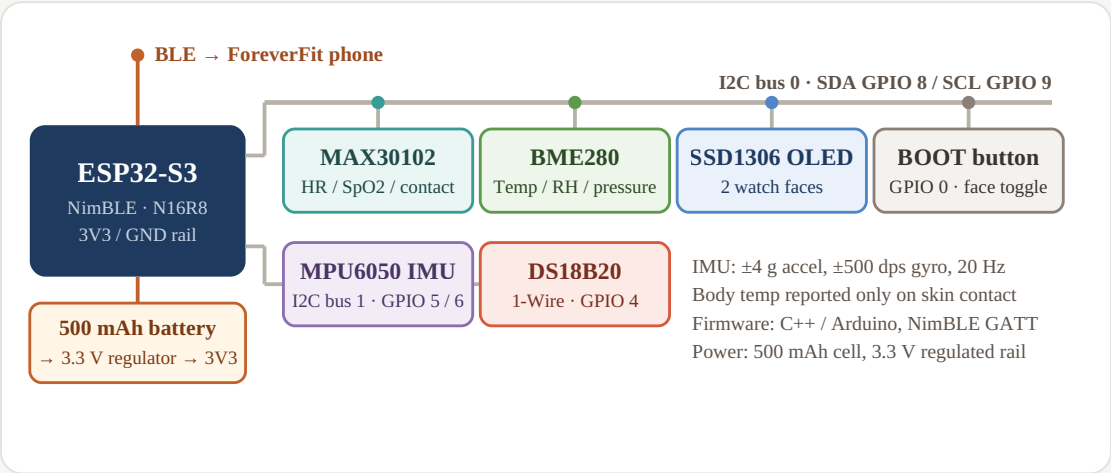
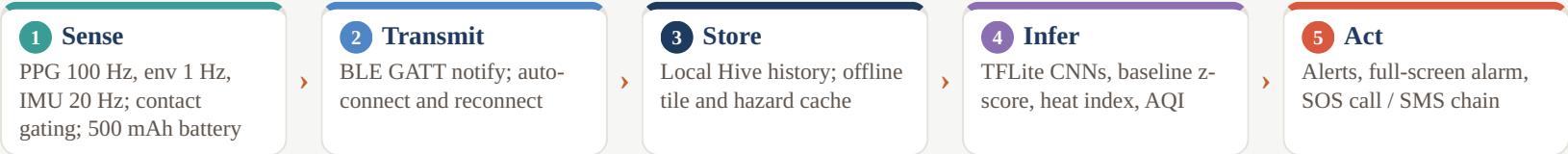
APP IN ACTION

Real screens from the live Android build on a physical phone.

More: wellness breakdown, AI assistant, health log, history charts, watch customisation, settings.

[View all app screens](#)

METHODOLOGY AND PROCESS FOR IMPLEMENTATION (pipeline, circuit, working prototype)



Working hardware: live battery-powered ForeverBand OLED face, time synced from the phone over BLE, BME280 readings

ON-DEVICE AI MODELS (all inference runs locally on the phone)

MODEL	INPUT	ARCHITECTURE / SIZE	TRAINING DATA	HELD-OUT RESULT
Fall detection (phone)	3 s accel + gyro window, 20 Hz	1D-CNN, ~14k params, ~60 KB TFLite	UMAFall, 19 subjects, 746 trials	92.0% precision / 92.0% recall at 0.80
Fall detection (watch)	Wrist IMU + phone IMU fused	1D-CNN, ~62 KB TFLite	UMAFall wrist + pocket	89.8% recall at 0.50 (recall-first)
Activity classifier	Phone IMU window	1D-CNN, ~60 KB TFLite	MotionSense	99.9% accuracy (still / walk / run)
AI health assistant	Text, voice, photo, PDF	Gemma 4 E2B, ~2.6 GB, CPU	Pretrained; grounded in local data	Fully offline, opt-in, Wi-Fi-only download

ANALYSIS OF THE FEASIBILITY OF THE IDEA

- **Technical:** working prototype on real hardware: battery-powered ESP32-S3 band ↔ BLE ↔ Android phone
- **Compute:** ~60 KB CNNs run in real time on a budget phone; LLM on CPU, no server
- **Operational:** no backend; free public APIs (Open-Meteo, USGS) with offline caches
- **Maturity:** ~16k lines of Dart, 54 unit tests, compile-checked firmware
- **Cost:** ≈ ₹2,200 prototype BOM (approx. retail): ESP32-S3 ₹650, MAX30102 ₹450, BME280 / MPU6050 / DS18B20 ₹550, battery + regulator ₹250, OLED + strap ₹300

PERFORMANCE, POWER AND RANGE

Power supply	500 mAh battery → 3.3 V regulator → shared 3V3 rail
Battery life	≈ 5 h streaming at ~100 mA (estimate; bench test pending)
BLE range	~10 m indoors (typical BLE); auto-reconnect
Fall-detection latency	≈ 0.5–1 s impact → alert (0.5 s inference, 2 hits), then 10 s cancel window

POTENTIAL CHALLENGES AND RISKS	STRATEGIES FOR OVERCOMING THESE CHALLENGES (applied in the prototype unless marked planned)
Fall false positives: a fast phone pickup can look like a short fall	Threshold swept on held-out data (0.80 is best), 2 consecutive windows required, 10 s "I'm OK" cancel, so a false alarm costs one tap and not a call
Android kills background sensing when the screen is off	Foreground service with a visible notification; full-screen escalation over the lock screen only for a real fall
OEM restrictions (MIUI / HyperOS) block scheduled alarms	Diagnosed with adb / dumpsys; reminders moved to a reliable in-app scheduler, plus a battery-optimisation exemption flow
Unreliable readings when the band is off the wrist, moving or still settling	Contact gating via the MAX30102 IR signal; HR / SpO2 only after a 2 s settle and 4 clean beats, outlier rejection, 5 s staleness reset; 1 min DS18B20 settle; "measuring" shown instead of a false value
Hardware variance: a rebadged IMU (MPU6500) is rejected by stock libraries	Root-caused by WHO_AM_I readback; replaced with a raw-register driver checked against gravity readings
Networks fail during disasters	Offline map tiles, a state-level hazard baseline (BIS / IMD), last-known GPS and on-device AI; SMS fallback if calls fail
Large LLM (2.6 GB) on low-end phones	Opt-in, Wi-Fi-only download with a size warning; CPU backend after a GPU crash was found; core safety never depends on it
Battery life on a small 500 mAh cell	1 Hz vitals / environment notifies and 1 Hz OLED refresh; planned: light sleep, longer BLE connection interval, OLED auto-off
BLE link drops when the phone is out of range	Auto-reconnect on scan and app resume; phone-side safety features keep running while disconnected
Alert fatigue from non-emergencies	Tiered alerts: full-screen siren only for imminent earthquakes and cyclones; weather changes stay as notifications with per-condition cooldowns

VIABILITY: ADOPTION CHANNELS AND SCALE-UP PATH (consumers, healthcare providers, disaster-response agencies, public health programs)

Consumers (B2C) Low-cost band plus a free app for elderly users, outdoor workers and chronic patients; no subscription needed	Healthcare providers (B2B) Remote monitoring of discharged or chronic patients; exportable history for consultations	Disaster response and public health (B2G) NDMA / SDMA heat-action plans, ASHA-worker bulk kits, flood and cyclone camps	Roadmap Custom PCB and enclosure, encrypted backups, OTA firmware, clinical validation of HR / SpO2, Indian-language UI
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POTENTIAL IMPACT ON THE TARGET AUDIENCE

VULNERABLE GROUP	RISK DURING DISASTERS AND DAILY LIFE	WHAT FOREVERFIT / FOREVERBAND DOES
Elderly citizens	Falls when alone, delayed help, cardiovascular events	Automatic fall detection, SOS call chain to family, spoken vitals summary for responders
Outdoor workers	Heat stress, dehydration during heat waves	Heat index from wearable and forecast, body-temperature warnings, hydration reminders
Chronic conditions	Respiratory and cardiac flare-ups, missed medicines	SpO2 and AQI alerts, personal-baseline HR anomalies, medication schedules, BP and glucose logs
Rural and low-connectivity	No internet or cloud; far from care	Offline-first: on-device AI, cached maps and hazards, offline LLM for health questions
Women	Health needs under-served by generic apps	Menstrual cycle tracking with prediction, in the same private record

MARKETING AND GO-TO-MARKET POSSIBILITIES

- **Direct to consumer:** Play Store / App Store freemium app; band sold on Amazon, Flipkart and D2C at ≈ ₹2,200 BOM
- **Government procurement:** GeM listing for SDMAs, municipal heat-action plans, NDMA / NHM pilots
- **Employers and CSR:** construction, mining, delivery and farm cooperatives for outdoor-worker safety
- **Healthcare partners:** clinics, pharmacies and elder-care homes bundling the band with remote monitoring
- **Insurers:** wellness-linked premium discounts for users who share summaries (opt-in only)
- **Community reach:** ASHA workers, NGOs and campus drives; seasonal campaigns before heat-wave and monsoon peaks
- **Platform:** open BLE protocol lets other bands and smartwatches plug in as a white-label SDK

BENEFITS OF THE SOLUTION (social, economic, environmental, etc.)

Social

Faster emergency response, reassurance for families, guidance with no internet or literacy barrier (voice input, read-aloud)

Economic

≈ ₹2,200 band, zero per-user cloud or AI cost; early warnings avoid costly emergency admissions

Environmental and disaster resilience

Turns heat-wave, AQI, flood and cyclone data into personal "is this dangerous for me now?" actions

Privacy and data sovereignty

Health data and AI chats never leave the device; the user controls export and sharing

Public health

Low-cost, scalable sensing layer for NDMA heat-action plans and community health programs

Accessibility

Works on the phone alone without the band; large-type UI; SOS in one tap

USE CASE: AN OUTDOOR WORKER ON A HEAT-WAVE DAY

1. Early warning

Forecast heat index reaches "danger"; the map banner and a notification fire

2. Prevention

Hydration reminders; the wellness score drops as risk rises

3. Detection

Elevated body temp with a dangerous heat index gives a heat-stress alert

4. Fall

The worker collapses; the wrist + phone fall
CNN alerts within about 1 s

5. Response

No "I'm OK" in 10 s: emergency call with spoken vitals and location, then the emergency contact, then SMS

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

- 1 **UMAFall: multisensor dataset for automatic fall detection**
Casilari E., Santoyo-Ramón J.A., Cano-García J.M., Procedia Computer Science 110 (2017). Trains and validates both fall CNNs.
figshare.com/articles/dataset/UMA_ADL_FALL_Dataset_zip/4214283
- 3 **Gemma open models and LiteRT on-device inference**
Google AI Edge. Runtime for the offline LLM assistant and the TensorFlow Lite CNNs.
ai.google.dev/gemma · ai.google.dev/edge/litert
- 5 **Heat-wave criteria and heat action plans**
India Meteorological Department heat-wave definitions; NDMA Guidelines on Prevention and Management of Heat Wave (2019).
mausam.imd.gov.in · ndma.gov.in
- 7 **USGS Earthquake Hazards Program feed**
Live seismic events (M4.0+ within 200 km) for the risk map and the imminent-disaster alarm.
earthquake.usgs.gov/fdsnws/event/1
- 9 **Body fat from BMI (Deurenberg formula)**
Deurenberg P., Weststrate J.A., Seidell J.C., British Journal of Nutrition 65 (1991) 105–114.
doi.org/10.1079/BJN19910073
- 2 **MotionSense: smartphone accelerometer and gyroscope activity dataset**
Malekzadeh M. et al., "Mobile Sensor Data Anonymization", IoTDI 2019. Trains the activity CNN.
github.com/mmalekzadeh/motion-sense
- 4 **The Heat Index Equation (Rothfus regression)**
Rothfus L.P., NWS Technical Attachment SR 90-23 (1990); NOAA / NWS heat index. Used for the heat-stress risk score.
wpc.ncep.noaa.gov/html/heatindex_equation.shtml
- 6 **Open-Meteo Weather and Air Quality APIs**
Free, keyless forecast, US AQI, PM2.5 and PM10 data, used for live hazard input with an offline cached fallback.
open-meteo.com
- 8 **BIS IS 1893 (Part 1):2016 seismic zoning**
Bureau of Indian Standards seismic zone map, used as the offline state-level hazard baseline.
bis.gov.in
- 10 **Sensor datasheets**
Analog Devices MAX30102, MAX31820 / DS18B20, Bosch BME280, InvenSense MPU-6050 / 6500, Espressif ESP32-S3.
analog.com · bosch-sensortec.com · espressif.com

OUR OWN RESEARCH: FALL-THRESHOLD SWEEP (held-out subjects 18–19, 1,504 windows)

THRESHOLD	PRECISION	RECALL	MISSED FALLS
0.50 (default)	88.2%	93.8%	7
0.70	90.4%	92.0%	9
0.80 (chosen)	92.0%	92.0%	9
0.90	91.4%	85.7%	16

- Impact-point labelling replaced whole-trial labels, which had inflated scores
- Two extra heuristics (free-fall duration, impact g-force) were **disproven** by real drop-test logs and removed

TEAM ABBOY AND SOURCE

Source: [GitLab](#) (ForeverFit + ForeverBand, full app, firmware and ML code)
Demo video: [YouTube](#) (hardware and app walkthrough)

Smarajit Datta · Full Stack

Ankit Gupta · Backend Design

Pradipta Bhattacharya, Saptak Chatterjee · Embedded Development

Abhijit, Anushka Mondal · Frontend Development

FINDINGS FROM ON-DEVICE TESTING (root-caused with logcat, adb and serial logs rather than guessed)

Sensor rate ≠ requested rate

Phones delivered IMU data at over 100 Hz against a 20 Hz request, which squeezed 3 s windows into 0.4 s; throttling restored the trained timing

Rebadged IMU silicon

The WHO_AM_I register returned 0x70 (MPU6500), not 0x68; a raw-register driver made the "MPU6050" usable

BLE auto-connect race

The scan call returns when scanning starts, not when it ends; waiting on the scan-state stream fixed auto-connect

Background sensing limits

Android stops sensor delivery to backgrounded apps, so a foreground service is required for 24×7 fall safety